Examiner reports

Q1.

The candidates found this question more demanding than the previous questions. In part (a), there were many good responses, but a number of candidates lost marks because they did not fully justify their final answer. When giving answers to questions like part (a), where an answer correct to three significant figures is required, candidates should be advised to write an answer to more than three significant figures before giving their final answer; otherwise examiners do not know whether the value has been calculated or copied from the question paper.

Part (b) was generally done well, sometimes using the answer given in part (a).

In part (c), there was quite a mixture of responses. A significant number of candidates thought that the distance would change. Of those who felt that the distance would not change, only some were able to justify this statement.

Part (d) was done well by some candidates, but one of the most common errors was to use a time of 3.13 seconds. A few candidates did not resolve the velocity and used 20 instead of 20 sin 50° in their calculations.

In part (e), candidates were simply expected to write down the velocity, but very few candidates did this. A significant number tried to calculate the velocity. In this case, there were often minor errors in the calculation of the speed. When stating the direction of the velocity, some candidates simply stated 50°, but did not specify that this was below the horizontal or draw a diagram to show this.

Q2.

The candidates found part (a) of this question relatively straightforward, but part (b) was found to be very demanding, and very few correct solutions were seen. Candidates lost marks in part (a) mainly due to errors in the resolving: for example using sin 30° instead of cos 30° in part (a)(ii). A few candidates lost one mark because they gave their final answer for the coefficient of friction as 0.31 rather than 0.313.

In part (b), very few candidates realised that they needed to express the normal reaction in terms of the tension. This error resulted in solutions that were flawed. Most of the candidates who did express the normal reaction in terms of the tension went on to gain full marks for this part of the question, although a few did make arithmetic or algebraic errors.

Q3.

Some candidates found it very difficult to form a correct quadratic equation based on the vertical motion of the pellet. One of the main difficulties was dealing with the sign of the number 3.

Many candidates did part (b) well, and some who had been unable to find the time were able to use the printed answer to calculate the range. Parts (c) and (d) were found more challenging by candidates. While there were a few very good solutions, many candidates did not realise how to obtain the required answers, especially in part (d).

Q4.

Generally part (a) was done well, whilst part (b) caused more difficulties. The fact that the acceleration was given did seem to help many candidates, but some did not justify the minus sign. Applying the constant acceleration equation to find the distance was also done well, although a few candidates took $u = 0$ and $v = 4$. There were a number of candidates who used two constant acceleration equations, finding the time taken to stop as an intermediate step. A few of these candidates did not go on to find the distance.

In part (b), many of the candidates were able to find the magnitude of the friction force. In a few cases the candidates did not show how the reaction force was obtained. Full marks were not awarded if the candidates simply stated $R = 1.84$ without any justification. When calculating the acceleration there were many sign errors in candidates' equations and some candidates also included a"4" which appeared as if it was a force. While many candidates realised that the puck would remain at rest, only a relatively small number of them were able to give a convincing explanation. Many of the explanations contained vague statements and lacked precision and clarity.

Q5.

Many candidates did well on this question, scoring a good number of marks. Part (a) was often answered well, but some candidates did include a number of spurious reasons. In part (b), there were many good responses. Some candidates did not show clearly how they obtained the equation which they went on to solve to find the time.

A large number of candidates found the time to the maximum height and doubled this value. Some of these candidates did not explain why this doubling had taken place and a 2 seemed to appear in their equations with no justification. Part (c) was generally done well. Some candidates gained full marks on part (c) without doing part (b). A few candidates had difficulty solving the inequalities to find the values of V.

Q6.

The candidates found this question more demanding, especially part (b). There were many good responses to part (a), but some candidates were not sure about how to go about finding the angle. Part (b) caused more difficulties. Some candidates were not able to form appropriate equations, and some of those who did made mistakes solving them. Some candidates found two times but did not find the difference between them.

Q7.

Parts (a) and (b) were generally done well. There were a few candidates who obtained the printed answer in part (a) from incorrect working. A popular approach was to find the time to the maximum height and to double this to find the time of flight. Part (b) was done very well, with some candidates who had not done part (a) using the printed answer to enable them to proceed.

Part (c) was more challenging and there were fewer correct completions. The main area of difficulty for this part of the question was the setting up of a quadratic equation. Many candidates included a 1 to allow for the change in height, but gave it the wrong sign. A number of candidates solved incorrect quadratic equations. A further issue concerned the solution of the quadratic. Some candidates showed both solutions and indicated that the solution required was the positive one; other candidates simply ignored the second solution without giving any justification for doing so.

In part (c), some candidates found the time by summing two times. For some of these students this approach worked well, but for others, especially when they involved the time to the highest point, there were quite a number of errors in their working.

Q8.

Parts (a), (b) and (c)(i) of this question were answered well by many candidates. However, some candidates lost one or more accuracy marks for parts (b) and (c)(i). To answer part (c)(i), most candidates evaluated 10 cos 26.9° and 10 cos 74.4° to arrive at the correct conclusions. A small number of candidates stated that the can will be knocked off the wall if 10 cos α > 8, or cos α > 0.8, and then thought that this would imply α > 36.9°.

Q9.

There were a fair number of good attempts at parts (a) and (b) of this question, but part (c) was found to be more demanding. In part (a), there were three main sources of error: introducing a sign error; using cos instead of sin; and not applying the quadratic equation formula correctly.

There were many good answers to part (b), with some candidates using the time given in part (a) when they had been unable to obtain it for themselves.

There were very few complete answers to part (c). Some candidates did not make any form of serious attempt at this part of the question. Some candidates only found one component of the velocity, but not the other; most often they found the vertical component, but not the horizontal component.

Q10.

Candidates seemed to either do very well or very badly on this question, with many producing completely correct solutions. Some candidates did part (a) by finding the angle α first and then using trigonometry to find F, rather than finding F first as suggested in the question. There were some candidates who were able to form the two equations, F cos α = 6 and F sin α = 5 but who then made no progress in solving these to find either F or α. A small number of candidates did not know how to start this question and produced no working at all.

Q11.

In part (a) there were many good diagrams, but also a number that included extra forces, especially parallel to the slope. On some diagrams the arrows or labels were missing. Part (b) was done well and the vast majority of candidates did show enough working to gain all of the marks. Some candidates clearly used the given answer to correct their working. Part (c) was more demanding. When finding the acceleration, the main errors were not dividing the resultant force by the mass; making a sign error to obtain a = –0.171; not considering both forces, to produce equations like 100a = 68.4; and not resolving the weight correctly.

In the last part, there were some good explanations, but there were also a lot that were not acceptable. Some candidates wrote about the resistance being related to the acceleration rather than the speed or velocity. A few also thought that the resistance would depend on the slope.

Q12.

Part (a) was usually done very well, but in some cases very badly. Many candidates produced good complete solutions. A few made arithmetic or algebraic errors that often resulted in the loss of the final accuracy mark. A few candidates found the time, but not the height. Part (b) was more demanding. Some candidates did not really make any progress with this part of the question. For those that got started, the most common errors were to include a sign error in the quadratic equation that they used to find the time; and to produce a correct quadratic equation, but to make an error in finding its solutions.

Some candidates did use a two-time approach, finding the time to go up to the maximum height and the time to come down to the tree. Many of these candidates were successful, but there were problems associated with getting the right distance to use to find the time down to the tree.

Q13.

This question caused problems for a number of candidates. The most common error, which created difficulties throughout the question, was to treat the ball as if it had an initial vertical component of velocity, often equal to 20 m s^{-1}. This would lead to equations of the form $2.45 = 4.9t^2 \pm 20t$. A few candidates who were unable to do part (a) were able to use the printed answer from part (a) to complete part (b) correctly, but some did include an initial vertical component of velocity again.

Part (c) of this question was more demanding. There were two main errors. The first was to work with the position of the ball when it hit the ground. These candidates would calculate an angle based on the two distances that they knew. A typical answer for these candidates was $\theta = \tan^{-1}\left(\frac{2.45}{14.14}\right)$. The other error was to only find one component of the ball's velocity.

Q14.

This question, particularly part (a), proved to be more demanding and was not answered well by the weaker candidates. In part (a), some candidates ignored the instruction to resolve horizontally. The main issue was that the vast majority of candidates assumed that the tensions were equal and did not assign variables such as T_1 and T_2 to the tensions. In parts (b) and (c), some candidates did not take account of the fact that there were two tensions to take into account. In addition, some candidates made resolving errors, using sin 30° instead of cos30°. In part (c), some candidates felt that there was no need to resolve the forces, although they had done so in part (b). Also a small number of candidates found the weight rather than the mass.

Q15.

Although there were many correct force diagrams, there were also a significant number which included friction or other forces as well as the two required. A good number of candidates were able to find the acceleration when the slope was smooth, although there were a few candidates who did not show enough working, simply giving statements like "$a = \frac{1}{2}g = 4.9$ m s^{-2}". In part (b), many candidates were able to find the acceleration successfully, but there were two common errors. The first was to assume that they could use = 5 v without any justification. The second was to simply divide the distance by the time. When finding the friction force, some candidates used $F = ma$ to obtain a friction force of 7.5 N. There were also a number of correct solutions. When finding the coefficient of friction, many candidates with incorrect values for the friction benefited from the follow through marks and gained full marks. There were however some cases where the normal reaction was incorrect either because no resolving had taken place or because the resolving had been incorrect.

In part (c), there were some good answers from candidates, but there were also quite a few which stated that the acceleration would be less, but did not give a good enough explanation for this.

Q16.

In part (a) the majority of the candidates gained full marks, although there were a small number of candidates with poor responses, such as “has no mass”. Parts (b) and (c) were also done very well, but with some minor errors present. There were fewer correct responses to part (d).

In a number of cases the speed was given correctly and the angle of 40° was also stated, but no indication that it was below the horizontal was included. A few candidates worked out the horizontal and vertical components of the velocity when the arrow hit the ground and then calculated the speed and the direction. This was a very hard way to earn the marks, but it did work for some of these candidates. Generally, part (d) was either done well or very badly with candidates not knowing how to start.

Q17.

The responses to part (a) were very good. The vast majority of candidates were familiar with the kinematics equations for the motion of a projectile.

Many candidates answered part (b)(i) of the question correctly. However, some candidates misunderstood the axes and substituted –1 instead of 1 for y. Evidently, because the answer was given, some of these candidates were able to rectify their mistakes but others gave wrong answers.

The candidates understood and used the condition for real solutions for the quadratic equation to arrive at the relationship requested for part (b)(ii). For part (b)(iii), almost all candidates substituted 5 for R. However, some candidates were unable to proceed further and solve the quadratic equation in u^2 in order to find the minimum speed of projection. There were other candidates who attempted to use differentiation to find the minimum speed. These candidates tried to minimize the expression $u^4 - 20u^2 - 2500$ rather than finding the minimum value of u.

Q18.

This question proved difficult, and a number of candidates showed a lack of appreciation of the differences between the horizontal and vertical components of the motion. Solutions in part (a) sometimes lacked convincing reasoning. Algebra let some down in part (c), and part (d) proved very discriminating on the understanding of the motion.

Q19.

Parts (a)(i), (ii), (v) and (b) proved routinely successful for well prepared and accurate candidates. Parts (a)(iii) and (iv) tested the understanding of the forces in the static situations and proved highly discriminating.

Q20.

This question proved quite challenging, with the least successful candidates confusing the horizontal and vertical components of motion, and in some cases trying to incorporate expressions for the greatest height, horizontal range, and time of flight, thereby making the question harder. Part (a) was mostly done well, but some assumed vertical motion was involved. Part (b) was sometimes omitted, while a significant number did not realise that the calculation could be done in one stage and wasted time in carrying out a series of calculations often losing an accuracy mark due to rounding values within their working. In part (c)(i) common errors included finding the vertical component of velocity only, using ‘24’ as the initial velocity with no component, and mixing horizontal/vertical motions or velocity/displacement in calculations. Those successful in part (c)(i) usually completed part (c)(ii).

Q21.

Part (a) The vast majority of candidates were familiar with the equations of motion for a projectile. They were able to eliminate t and arrive at the required equation of the trajectory.

Part (b) The algebraic manipulation and ‘tidying up’ of the quadratic equation in x and the subsequent task of solving it were prone to errors for some candidates. Some candidates who solved the equation correctly did not identify the required answer from their two solutions.

Part (c) Most candidates stated the two assumptions; no air resistance and the ball is a particle. A small minority of the candidates mistakenly stated that the ball was a particle without that mass.

Q22.

There was a good number of correct force diagrams. Some candidates added one or both of the components of the tension to their force diagram. In some cases these appeared as if they were additional forces. Parts (b) and (c) were done well by most candidates who seemed to have benefitted from the printed answers. There were some candidates who did not gain full marks, because they did not show sufficient working. In part (d), many candidates obtained the correct value. Some candidates used the two printed answers as expected, if they had not been able to obtain the two values for themselves. There were some very good answers to part (e), but there were also quite a few confused ones as well, with many candidates finding it hard to give a good answer.

Q23.

Part (a) of this question was done well by several candidates, while part (b) was found to be more demanding. In part (a), the candidates seemed to have been helped by the printed answer. In part (b), the major problem was setting up an appropriate equation to find the time of flight. Those who could do this generally went on to complete the question correctly. One issue that was evident was that a small number of candidates did not resolve the velocity into components, but simply worked with 30 wherever they needed a velocity.

Q24.

Part (a) was generally answered well. There were however some curious assumptions, for example “the ball has no mass”, which were not appropriate. Part (b) was generally done quite well, with candidates obtaining the results without too much difficulty. Part (c) proved to be more demanding, but candidates often tended to do very badly or very well. If they could produce a correct equation for t they could often gain all the marks, but other candidates could not see how to form an equation that would enable them to start the question. In a few cases the candidates rounded the time of flight and then obtained an incorrect value for the range.

Q25.

A variety of force diagrams were seen on the candidates’ scripts. Some were very good, but errors included the omission of forces and showing the friction force acting up the slope. In part (b) the candidates were clearly helped by the printed answer and there were many correct responses. In some cases the candidates did not show enough working to gain full marks. Part (c) proved to be very challenging, with only a very small number of candidates getting the correct coefficient of friction. The main problem was that the candidates did not include all of the terms that were needed in their equations of motion.

Q26.

The vast majority of the candidates answered part (a) of the question correctly.

For part (b), almost all candidates substituted 100 for x and 4 for y. However, some candidates had difficulty simplifying the resulting equation.

For part (c), most candidates stated that they assumed the ball was a particle, there was no air resistance or that the only force acting on the ball was the Earth’s gravitational force. However, there were some candidates who stated the assumption of the ball being a particle without mass!

Q27.

Diagrams were mostly very good with only occasional lapses in the direction of the normal reaction force or in poor labelling of forces. Part (a) proved popular and successful for most candidates. In part (b)(i) a number failed to appreciate that the forces involved included the weight component, but many went on to be successful in part (ii).

Q28.

This question proved more discriminating than expected.

Q29.

Many candidates were successful in this question, while others appeared to have little appreciation of the directions of forces and relevant components. Errors with signs and selection of trigonometrical function were relatively frequent.

Q30.

In part (a) diagrams were often disappointing, with common errors including an additional force in the direction of motion and the inclusion of acceleration or velocity as a force.

Parts (b) and (c) were done quite well, but a significant number of candidates selected the wrong component when finding the required forces and some failed to give their final answer to the required degree of accuracy stated in the rubric.

Q31.

Many candidates were fully successful in part (a). In part (b), however, complete equations of motion were rare and components were often inappropriately applied: either incorrect or used with inappropriate forces.

Q32.

The significance of the direction of the resultant force was lost on many. Again answers were often disappointing, with errors finding components and confusion with directions of forces.

Q33.

Many candidates attempted this question satisfactorily, realising that the sum of the components of the three given forces was zero parallel to Ox, but non-zero parallel to Oy. Resulting equations were often fully correct and accurate arithmetical work followed. Some errors in selecting the correct component or sign were seen, but the most common mistake was failure to find the magnitude of the resultant and thereby complete part (b).

Q34.

A surprising number of candidates failed to score both marks for the forces diagram in part (a), with the most common errors being the omission of the friction force or stating the normal reaction force to be equal to the weight. Parts (b) and (c) were done well, and many were successful in part (d), although some omitted a force from their calculations.

Q35.

This last question revealed an improved understanding of projectile motion, as well as clear and accurate arithmetic work. A variety of methods were used in all parts of the question, including use of the symmetry of the motion.

Part (a) was done well, with vertical motion being considered both upwards and downwards. In part (b), again both directions of the vertical motion were used. Part (i) was done well, but part (ii) proved more discriminating, with a weakness in solving quadratic equations being shown by some. Part (iii) was again mostly successfully and accurately completed.